THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET A

Chapter 4: **Carbon and its Compounds** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of covalent bonds in a molecule of methane (CH_4) is: 1

(a) 2 **(b)** 4 **(c)** 1 **(d)** 8

2. The functional group present in ethanol is: 1

(a) –OH **(b)** –COOH **(c)** –CHO **(d)** C=O

3. The IUPAC name of CH_3COOH is: 1

(a) methanoic acid **(b)** ethanal **(c)** ethanol **(d)** ethanoic acid

4. A student states: “Graphite and diamond are made of different elements.” The correct response is: 1

(a) Correct

(b) Incorrect — both are allotropes of carbon

(c) Correct only for graphite

(d) Cannot be determined

5. For the general formula C_nH_{2n}, Student A says it represents alkenes; Student B says alkynes. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

6. Which of these is a saturated hydrocarbon? 1

(a) C_3H_8 **(b)** C_3H_6 **(c)** C_2H_2 **(d)** C_2H_4

7. The addition of hydrogen to vegetable oils in the presence of nickel is called: 1

(a) hydrogenation **(b)** saponification **(c)** substitution **(d)** oxidation

8. The reaction of an acid with an alcohol to form a sweet-smelling substance is called: 1

(a) saponification **(b)** combustion **(c)** esterification **(d)** hydrogenation

9. Alkaline $KMnO_4$ converts ethanol into: 1

(a) ethene **(b)** ethane **(c)** ethanoic acid **(d)** methanol

10. The property of carbon to form long chains by bonding with other carbon atoms is: 1

(a) tetravalency **(b)** catenation **(c)** isomerism **(d)** allotropy

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why does carbon form covalent bonds rather than ionic bonds? 2

12. Draw the electron-dot structure of a nitrogen molecule (N_2). 2

13. What are micelles? 2

14. A student says that soaps work well in hard water. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What is a homologous series? Give three characteristics, with the example of alkanes. 3

16. Write equations for 3
(i) combustion of methane
(ii) chlorination of methane in sunlight
(iii) hydrogenation of ethene.

17. What are isomers? Draw the structures of the two isomers of butane. 3

OR

How would you distinguish between ethanol and ethanoic acid experimentally? Give two tests. 3

18. Explain the cleansing action of soap with a diagram of a micelle. 3

OR

Give reasons: 3
(i) Saturated hydrocarbons burn with a blue flame.
(ii) Unsaturated hydrocarbons burn with a sooty flame.
(iii) Air holes of a gas stove are kept open.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What are the two properties of carbon that lead to a large number of carbon compounds? Explain each. 5
(ii) Draw the structure of ethanoic acid.
(iii) Name the functional groups in propanal and propanone.

OR

(i) Describe the esterification reaction with an equation. 5
(ii) What is saponification? Write its equation.
(iii) Give one use of esters.

20. **Case study:** Compound X (C_2H_6O) is used as an antiseptic and in cough syrups. On heating with conc. H_2SO_4 at 443 K it gives Y, which decolourises bromine water. 5
(i) Identify X and Y.
(ii) Write the equation.
(iii) What is the role of conc. H_2SO_4?
(iv) Write the reaction of X with sodium.

OR

(i) Differentiate between soaps and detergents (any three points). 5
(ii) Why do detergents cause water pollution?
(iii) What is hard water?

END OF QUESTION PAPER — **Total: 40 Marks**